

# EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · Avijeeet Singh — Vortex AI · paper doi:10.5281/zenodo.20706893 (v0.1.0; current method: whitepaper v3, emem 213e273) · measurements 29–30 Sep 2026 · scripts: github.com/Vortex-AI/esa\_poster

## Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?



### 2 · FULL TRACE

#### From one token to the source pixel, without emem software

|                                                                                                                                                                                                                   |                         |                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A 698-line script (Python stdlib, blake3, cbor2, pynacl) checked <b>15 links in 17.9 s, all verified</b> . Grey: what must still be trusted after each check.                                                     |                         |                                                                                                                                                                                                 |
| 1–3                                                                                                                                                                                                               | <b>bytes → name</b>     | 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: nothing                                                                                               |
| 4–5                                                                                                                                                                                                               | <b>cell binding</b>     | cell inside the hashed body = cell in the token; cell decodes to the lat/lng the derivation used · nothing                                                                                      |
| 6–7                                                                                                                                                                                                               | <b>derivation</b>       | scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order is read from emem’s code; no published schema                                            |
| 8–9                                                                                                                                                                                                               | <b>upstream</b>         | Planetary Computer STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum; the pixel is the one containing the point (GDAL floor) |
| 10–11                                                                                                                                                                                                             | <b>server signature</b> | emem.dev’s ed25519 receipt over the request, cells and returned fact_cids · proves what emem.dev served, not that it is true                                                                    |
| 12–14                                                                                                                                                                                                             | <b>public log</b>       | entry 2,457,078 is included in the 2,556,451-entry log, whose signed root is consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI                                        |
| 15                                                                                                                                                                                                                | <b>key → domain</b>     | the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC                                                                                            |
| The whole proof fits in a 4,906-byte file that re-verifies with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it. research/repro/v8/trace_fact.py · verify_bundle.py · §22 |                         |                                                                                                                                                                                                 |

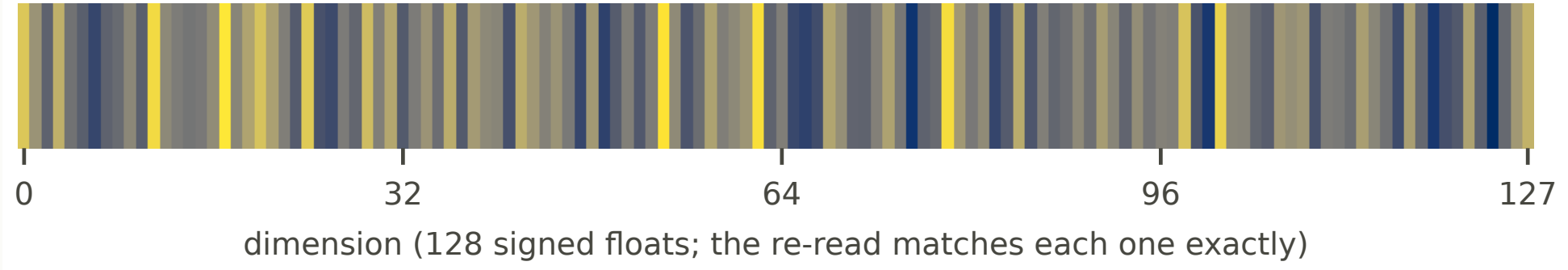
### 4 · ONE TOKEN, MANY CLIENTS

#### Ten client paths, one record

| PATH                                                                                                                                                                                                                               | CID       | RE-HASH         | SERVER SIG. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------|-------------|
| REST · MCP (Streamable HTTP) · A2A JSON-RPC                                                                                                                                                                                        | oj5ceccci | <b>3/3</b>      | <b>3/3</b>  |
| Python SDK · TypeScript SDK                                                                                                                                                                                                        | oj5ceccci | <b>2/2</b>      | <b>2/2</b>  |
| official MCP SDKs, LangChain MCP adapters, LlamaIndex tool                                                                                                                                                                         | oj5ceccci | <b>4/4</b>      | 3/3 †       |
| stock blake3 + cbor2 on the bytes                                                                                                                                                                                                  | oj5ceccci | <b>1/1</b>      | —           |
| same bytes from an untrusted mirror; tampered copy                                                                                                                                                                                 | j47f4xb3  | <b>rejected</b> |             |
| 3 repetitions each, 30 Sep 2026; value 0.4708994708994709 on every path. † the LlamaIndex tool drops the receipt. One responder (emem.dev) behind all paths: this shows client and protocol independence, not independent servers. |           |                 |             |

### 7 · FOUNDATION-MODEL EMBEDDINGS

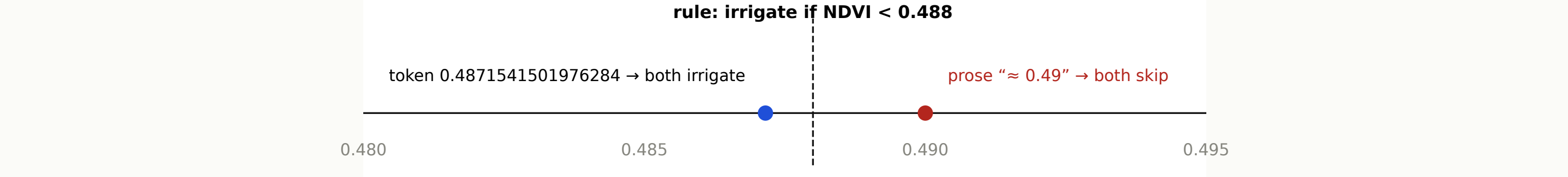
#### A TESSERA vector, re-read bit-exact by anyone



At the same cell, emem signs the 128-D **TESSERA v1** (Cambridge; Sentinel-1 + Sentinel-2, 2024) embedding as a model\_output record §9. Two range reads of the public product on Source Cooperative (128 B int8 + 4 B scale) give all 128 floats back with max |Δ| = 0; 4 of 4 facts tested match. emem re-serves this product; it does not run the encoder, and the record says so (tamper\_evidence: attester\_only: only the signer vouches for the encoder step). Encoders emem did run, such as Clay v1.5 §10, record the BLAKE2b hash of the weights used.

### 10 · WHY A PARAPHRASE IS NOT A CITATION

#### After a shared summary, 0 of 72 answers were right, yet 3 of 36 pairs agreed



| PRE-REGISTERED, GEMMA-4-12B + QWEN2.5-7B, ONE HOST                                                                                                                              | CORRECT      | AGREE        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| full context (control)                                                                                                                                                          | <b>72/72</b> | <b>36/36</b> |
| after a shared context summary (compaction)                                                                                                                                     | <b>0/72</b>  | <b>3/36</b>  |
| wrong-but-agreeing pairs (3/36) exceed correct answers (0/72): Fisher one-sided p = 0.035 §17; handoff n = 20 per arm: prose 2, dense retrieval 8, BM25 20, token bundle 20 §18 |              |              |

The trap value 0.4871541501976284 §21 is the value as signed on 17 Jul. It is a pre-fix read of the pixel south (section 3); the experiment tests fidelity to the cited value, not the ground. BM25 ties the bundle on short handoffs. Related work: telephone-game drift (Perez et al., ICLR 2025), “Lost in Compaction” (arXiv 2608.11242), pointer-based compaction ARC (arXiv 2607.25066).

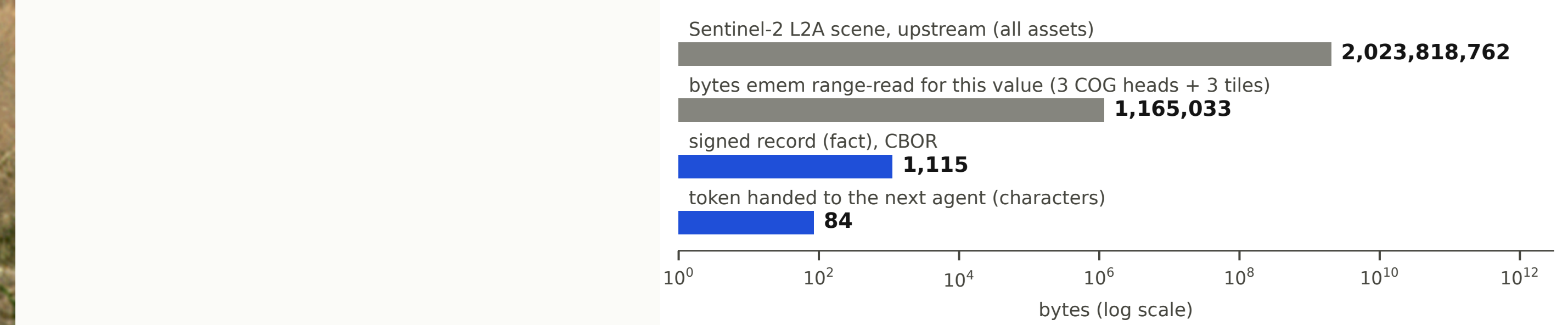
### 12 · EMEMIFY YOUR WORK

#### Five steps to sign and cite your own values, each run on 29–30 Sep

- 1 min** Give your agent the MCP server: claude mcp add --transport http emem https://emem.dev/mcp (18 core tools, 114 in all).
  - 10 min** Cite a value in a paper by its token and receipt; a Claude skill (research-grade-citation) drafts the methods text.
  - 1 hour** Register your own result, signed by your key: an NDVI change of −0.1298 (25 Sep 2025 → 2026) at this cell §11; emem recomputed it: bit-identical.
  - 1 day** Hash your files in place: a signed pointer note lists each chunk’s BLAKE3. This board is one §1.
  - 1 week** Run your own node: docker run ghcr.io/vortex-ai/emem. A local node re-read this pixel: bit-identical NDVI, a different fact\_cid (its own key signs).
- Apache-2.0. pip install ememdev · npm i @vortexai/emem · agent-to-agent (A2A) card at emem.dev/well-known/agent-card.json, signed.

### 1 · FROM SCENE TO TOKEN

#### 1.17 MB read from a 2.02 GB scene; the agent receives 84 characters



To sign one NDVI value, emem range-reads **0.058 %** of the scene: three 64 KiB COG headers and one tile each of B04, B08 and SCL. It stores the 1,115-byte record, not the tiles. The next agent receives 84 characters (46 GPT-4 tokenizer tokens).

We hashed the Earth Search (AWS) copies of the B08 and B04 COGs into two signed pointer notes §2 §3: all 292 tiles of each (238.7 MB, 232.9 MB) under one Merkle root; the files stay on AWS. The tile holding the cell is emem:tree:khigtqddb6jponqn4gv721f7e#row=108 (row = leaf index); its 1,586,895 bytes hash to that row. These are not the Planetary Computer file the NDVI was read from (281.9 MB, offset not pre-subtracted), and emem records no hash of that file.

```
$4 emem:fact:defi.zb572.xoso.zblec:oj5ceccile62uvmhedpuk67cjt2pqc7z33mtsuyffgakxbxmyaa
band indices.ndvi      day 20721 since 1970-01-01 = 25 Sep 2026
scene S2A_MSIL2A_20260925T054251_R005_143SFS · scene class 4, vegetation
DN      B08 3502  B04 1900  B0A offset -1000 (PC, baseline ± 04.00)
value 0.4708994708994709      (B08-B04)/(B08+B04) after offset: bit-identical
signer 777er3y1.. signed_at 2026-09-28T09:06:56Z · 1,115 B · BLAKE3 → oj5cecci..
```

Earth Search serves the same tile with the offset already subtracted (B08 2502, B04 900); NDVI is identical. Sizes: Planetary Computer STAC asset sizes and HTTP ranges logged 30 Sep.

### 3 · WHAT THE TRACE FOUND

#### Before 28 Sep, 162 of 200 sampled records carry a neighbouring pixel



Re-reading the COGs named inside the signed records showed that the 23 Sep record §7 carries the DNs of the pixel **10 m south** (dashed red: 2993/1972, SCL 5 bare soil, NDVI **0.3444**). The pixel that contains the point (blue) reads 0.4860. emem had rounded the fractional pixel index; GDAL takes the floor. The fix shipped on 28 Sep §8.

|                                                                |                             |
|----------------------------------------------------------------|-----------------------------|
| AUDIT OF PRE-FIX SENTINEL-2 RECORDS                            |                             |
| random sample, 200 records across 164 scenes and 19 bands      | <b>200</b>                  |
| DNs match the rounded (neighbour) pixel, not the floor pixel   | <b>162 · 81 % [75–86 %]</b> |
| rounding and floor give the same pixel                         | <b>38</b>                   |
| DNs match the floor pixel where the two rules differ           | <b>0</b>                    |
| ANDVI in emem’s signed change record at this cell, 23 → 25 Sep | <b>+0.127</b>               |
| ANDVI at the containing pixel                                  | <b>−0.015</b>               |

Re-reading the pixel found the error; the signatures date it. No signed byte was rewritten: old tokens still resolve to exactly what was signed, so the error is reproducible and datable. Post-fix records (after 28 Sep) read the containing pixel. Wilson 95 % interval. Figure: ● query point, blue = pixel containing it, dashed red = pixel emem read. The offset is a reproducibility error, not a claim about geolocation accuracy.

### 5 · TWO LLMS, EMEM AS THE ONLY MCP SERVER

#### Pre-registered handoff: forged tokens declined 5/5

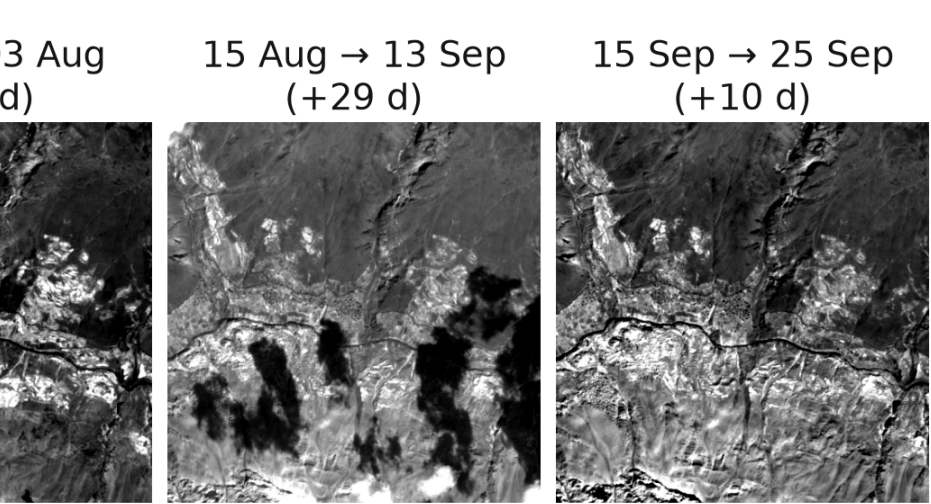
| B RECEIVES FROM A         | B CORRECT | B CHECKED                                    |
|---------------------------|-----------|----------------------------------------------|
| A’s token                 |           | <b>10/10</b> resolve, re-hash, receipt 10/10 |
| A’s own prose (16 digits) |           | <b>10/10</b> nothing to check                |
| prose rounded to “0.47”   |           | <b>0/5</b> —                                 |
| token with a forged cell  |           | <b>5/5 declined</b> 409 caught               |

A: Claude Sonnet 5.5 with emem as its only MCP server returned the exact token in 10/10 runs (13 s, 3.3 tool calls each). B: Claude Haiku 4.5. Rule: irrigate if NDVI ≤ 0.4705, a threshold **set on purpose** 0.0004 below the value so that 2-decimal rounding flips it; it shows the mechanism, not a rate. Token vs rounded prose: Fisher p = 0.0003; token vs full prose: a tie.

Pre-registration hashed before trial 1 (research/repo/data/v8/prereg.md). One model family for both agents; the local Qwen arm did not finish.

### 8 · RASTERS THROUGH TIME

#### A five-date B08 series; each member names its scene



An emem : cube : token §6 signs a list of five B08 raster records; each member states the date asked for and the scene used (+10 to +29 days later). The 13 Sep member is cloudy: scenes are chosen by scene-level cloud cover (≤ 40 %, relaxed in tiers), not a pixel mask.

### 11 · INTEGRITY ELSEWHERE, FOR THE SAME SCENE

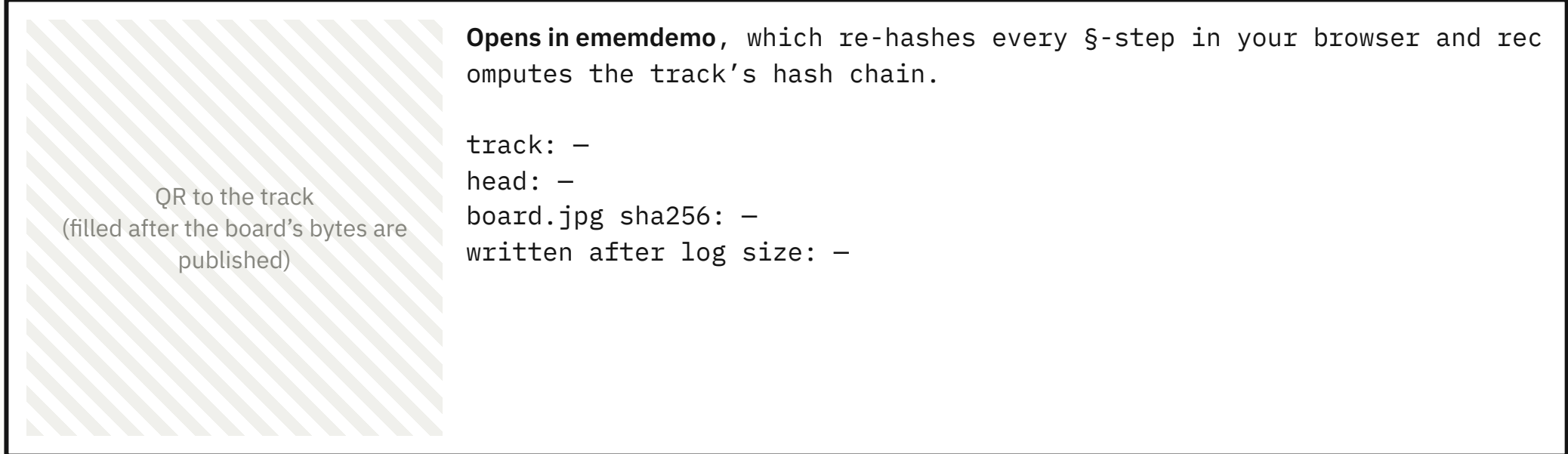
#### What an agent can check today, by source

| WHAT THE AGENT IS HANDED                      | INTEGRITY IT CARRIES                                                        |
|-----------------------------------------------|-----------------------------------------------------------------------------|
| Planetary Computer: 281,898,500 B B08 COG URL | opaque ETag, no checksum                                                    |
| Copernicus Data Space: 1,187,971,637 B SAFE   | MD5 + BLAKE3 of the whole product, unsigned                                 |
| NASA Earthdata MCP: a granule concept id      | catalogue revision; older revisions return 404                              |
| MCP tool results in general (spec 2026-07-28) | none; proposed in issue #3354                                               |
| <b>emem: one value</b>                        | <b>BLAKE3 name of the 1,115-byte record, ed25519 receipt, log inclusion</b> |

STAC and COG locate and read data; openEO and Earth Engine compute; emem names the exact value an agent cited so another agent can check it. emem does not yet store a hash of the upstream bytes (Source. hash is empty), so the upstream read is checked by re-reading the pixel, as in section 2.

### 14 · SEAL

#### Check this board



A file cannot contain its own hash, so this box is the only part of the board not inside the published bytes §1. Signed by the poster key njedkgit...; scripts to re-check everything without the viewer: research/repro/v8/.